Edge Computing Architectures: Paradigms, Platforms, and Strategic Deployment

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Abstract

Edge computing relocates compute and storage from centralised cloud data centres to resources near the data source, addressing the latency, bandwidth, and regulatory limits that hyperscale cloud alone cannot satisfy [1], [2]. This article surveys the architectural foundations of edge computing, compares the three dominant paradigms — cloudlets, fog computing, and Multi-access Edge Computing (MEC) — and examines modern cloud-native platforms, with a focus on lightweight Kubernetes distributions and serverless at the edge. It further treats edge AI, federated learning, security under zero-trust assumptions, and representative application domains such as autonomous driving, smart manufacturing, and healthcare. Recent 2022–2025 surveys and empirical benchmarks are emphasised. The article concludes that the three classical paradigms are converging into a unified end–edge–cloud continuum, and that the 2022–2026 research frontier has shifted from defining the edge to operating it at scale.

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1. Introduction

The global volume of machine-generated data is growing faster than the capacity of backbone networks to carry it. Gartner estimated that by 2025 roughly 75 % of enterprise-generated data will be created and processed outside traditional data centres or cloud regions, compared with about 10 % in 2018 [34]. This data-gravity problem — the fact that large datasets are expensive and slow to move — has motivated a structural shift in IT architecture: rather than transporting every byte to a hyperscale cloud, organisations push compute toward the data [1], [2], [12].

Edge computing is the resulting distributed paradigm. Shi *et al.* defined it as "enabling technologies that allow computation to be performed at the edge of the network, on downstream data on behalf of cloud services and upstream data on behalf of IoT services" [1]. Satyanarayanan later enumerated four motivations — latency, bandwidth, privacy as an "exposure firewall," and survivability — that remain the canonical justification for edge deployment [2].

This article is organised as follows. Section 2 traces the evolution from centralised cloud to the edge and introduces the data-gravity principle. Section 3 presents the three foundational paradigms (cloudlets, fog, MEC) and the end–edge–cloud continuum. Section 4 surveys modern platforms — lightweight Kubernetes and serverless at the edge — with empirical benchmarks. Sections 5, 6, and 7 cover edge AI and federated learning, security under zero trust, and

representative use cases respectively. Section 8 offers a future outlook, and Section 9 concludes.

Figure 1. Article roadmap.

2. From Cloud to the Edge

Computing has oscillated between centralisation and distribution for six decades: the mainframe era (1960s–70s), personal computing (1980s), client–server on the internet (1990s), and the cloud re-centralisation (2006–2015). The latter exploited economies of scale but imposed round-trip latencies of tens to hundreds of milliseconds — unacceptable for autonomous vehicles, remote robotic surgery, and immersive AR/VR [1], [2], [28].

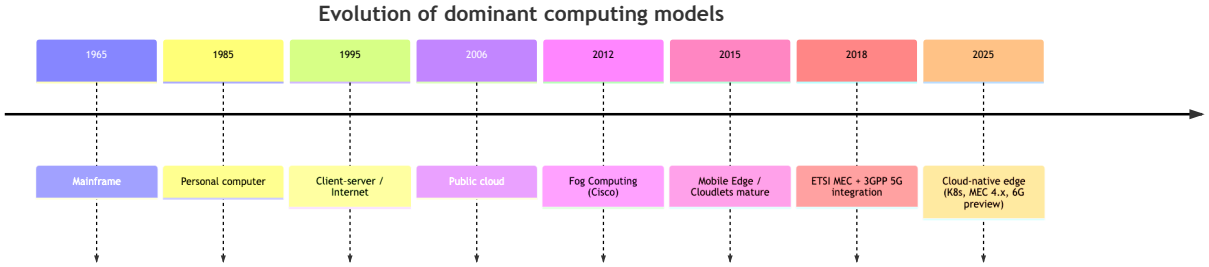

Figure 2. Timeline of dominant computing paradigms [1]–[6].

The 2012–2017 window is pivotal. Bonomi *et al.* coined *fog computing* as an IoT-oriented cloud-to-thing continuum [4]. Satyanarayanan *et al.* had already introduced *cloudlets* as trusted, resource-rich middle-tier nodes one wireless hop from the mobile device [3]. ETSI's Industry Specification Group (ISG) on Multi-access Edge Computing standardised MEC as the telecommunications-native flavour of the edge, explicitly co-designed with 5G New Radio [7], [9]. The 2020s then re-framed these three paradigms as deployment *flavours* of a shared end–edge–cloud continuum rather than competing alternatives [2], [10].

3. Architectural Paradigms

3.1 Cloudlets

A cloudlet is a small data centre or cluster "well-connected to the Internet and available for use by nearby mobile devices" [3]. Cloudlets trade hyperscale elasticity for proximity: a single wireless hop separates client from server, enabling crisp interactive response for mobile augmented reality, wearables, and latency-sensitive offload tasks [3]. Cloudlets originated the virtual-machine synthesis model, later generalised to container-based deployment.

3.2 Fog Computing

Fog computing, codified by the OpenFog Consortium and later standardised as **IEEE 1934-2018**, is a hierarchical, multi-layer architecture in which data, compute, storage, and applications are distributed across tiers between endpoint and cloud [5], [6]. The reference architecture enumerates eight pillars — *security, scalability, openness, autonomy, reliability-availability-serviceability (RAS), agility, hierarchy, and programmability* [5]. Fog excels in large-scale industrial IoT where thousands of sensors must be managed across complex facilities with east–west communication between peer nodes [29].

3.3 Multi-access Edge Computing (MEC)

MEC, standardised by ETSI ISG MEC since 2014, places compute inside or adjacent to the Radio Access Network [7], [9]. GS MEC 003 V4.1.1 (2025) defines the MEC host, MEC platform, MEC orchestrator, and the reference points (Mp1–Mp3, Mm1–Mm9, Mx1–Mx2) that bind them together [7]. From Release 17 onward, 3GPP integrates MEC into the 5G core through entities such as the Edge Application Server Discovery Function (EASDF) and the Uplink Classifier / Branching Point, enabling session-level traffic steering to local edge applications [32]. MEC is the natural host for V2X, cooperative perception, and ultra-reliable low-latency communication (URLLC) workloads [8], [9], [28].

3.4 The End–Edge–Cloud Continuum

Modern deployments treat the three paradigms as complementary layers. Ren *et al.*'s comprehensive survey synthesises them into a four-tier end–edge–cloud architecture and is the highest-value single reference for the comparison [10].

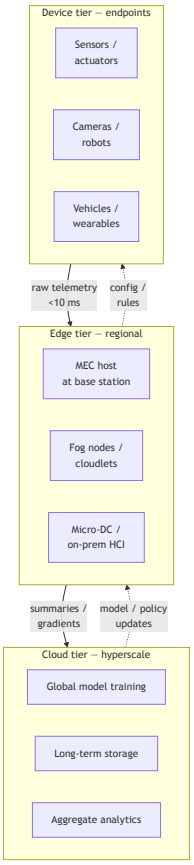

Figure 3. End–edge–cloud continuum with typical data flows [2], [10].

Table 1. Comparison of the three dominant paradigms, synthesised from [6], [10], [11], [22].

Dimension	Cloudlet (CMU)	Fog (Cisco / OpenFog)	MEC (ETSI / 3GPP)
Deployment	Small DC, one wireless hop	Hierarchical device→gateway→regional	Inside RAN / base station / telco edge
Owner	Local business, campus	Enterprise / utility	Mobile network operator
Connectivity	Wi-Fi / WLAN	Any (WLAN, LTE, industrial bus)	Cellular (4G/5G/6G)
Latency target	10 – 50 ms	10 – 100 ms	**1 – 10 ms**
Virtualisation	VM synthesis (originally)	Containers, VMs	NFV / containers standardised by ETSI
Standardisation	Academic / OpenEdge	IEEE 1934 (OpenFog)	ETSI ISG MEC, 3GPP Rel-17/18
Mobility	Hand-off between cloudlets	Limited	Native (follow-me workloads, EAS re-sel.)
Best-fit use cases	AR/MAR, wearables, mobile offloading	IIoT, smart grid, smart city	V2X/ADAS, immersive media, slicing

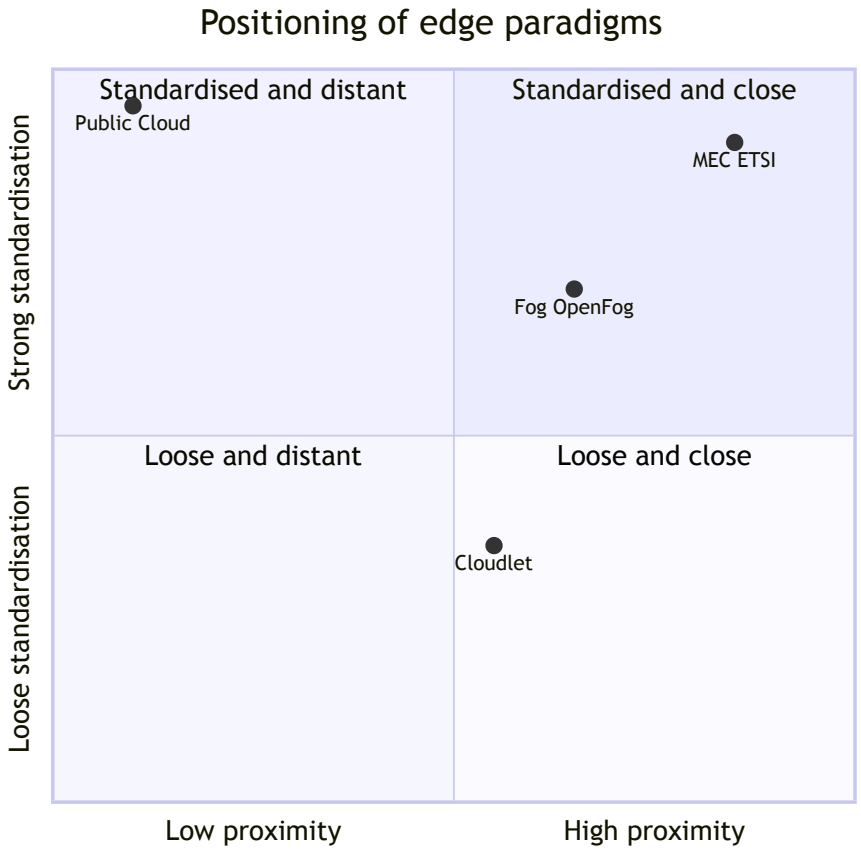

Figure 4. Paradigm positioning along proximity × standardisation axes.

4. Modern Edge Platforms

4.1 Kubernetes at the Edge

Kubernetes has become the *de facto* orchestrator for cloud-native workloads, but the upstream distribution is too heavy for constrained edge nodes. This has driven a family of **lightweight Kubernetes distributions** [13], [15].

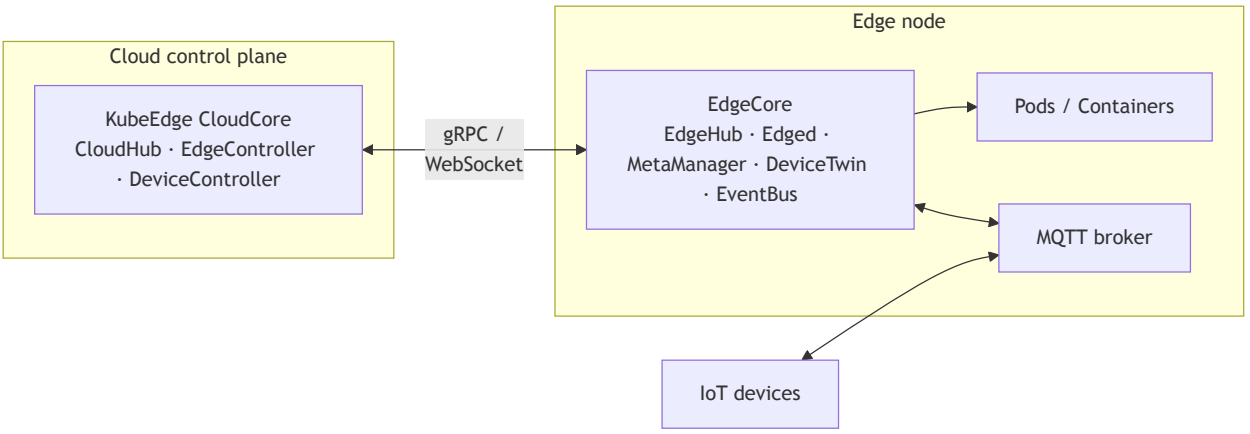

Figure 5. KubeEdge cloud–edge synchronisation architecture [14].

K3s, K0s, MicroK8s, KubeEdge, and OpenYurt each take a different path to reducing footprint [13], [15], [16]. Yakubov & Hästbacka's 2025 benchmark on Intel NUC and Raspberry Pi hardware found K3s to have the lowest resource footprint, K0s the strongest data-plane throughput, and KubeEdge the richest feature set for disconnected operation — but at a higher memory and disk-I/O cost that can wear consumer-grade SD storage [13]. Vaño *et al.*'s 2023 systematic review reached compatible conclusions across a wider feature matrix including Akri device management [15].

Table 2. Lightweight Kubernetes distributions for edge deployment (from [13], [15], [16]).

Distribution	RAM footprint	Packaging	Disconnected-edge support	Notable trait
K3s	< 512 MB	Single binary	Limited	SQLite default DB; CNCF-certified
K0s	Minimal	Single binary	Limited	Best data-plane throughput
MicroK8s	~238 MB worker	Snap (Ubuntu)	Moderate	Rich addon system
KubeEdge	Higher (split)	CloudCore + EdgeCore	**Native**	Device twin, MQTT, offline pods
OpenYurt	Moderate	Non-intrusive to K8s	Good (YurtHub)	Zero delta to upstream kubelet

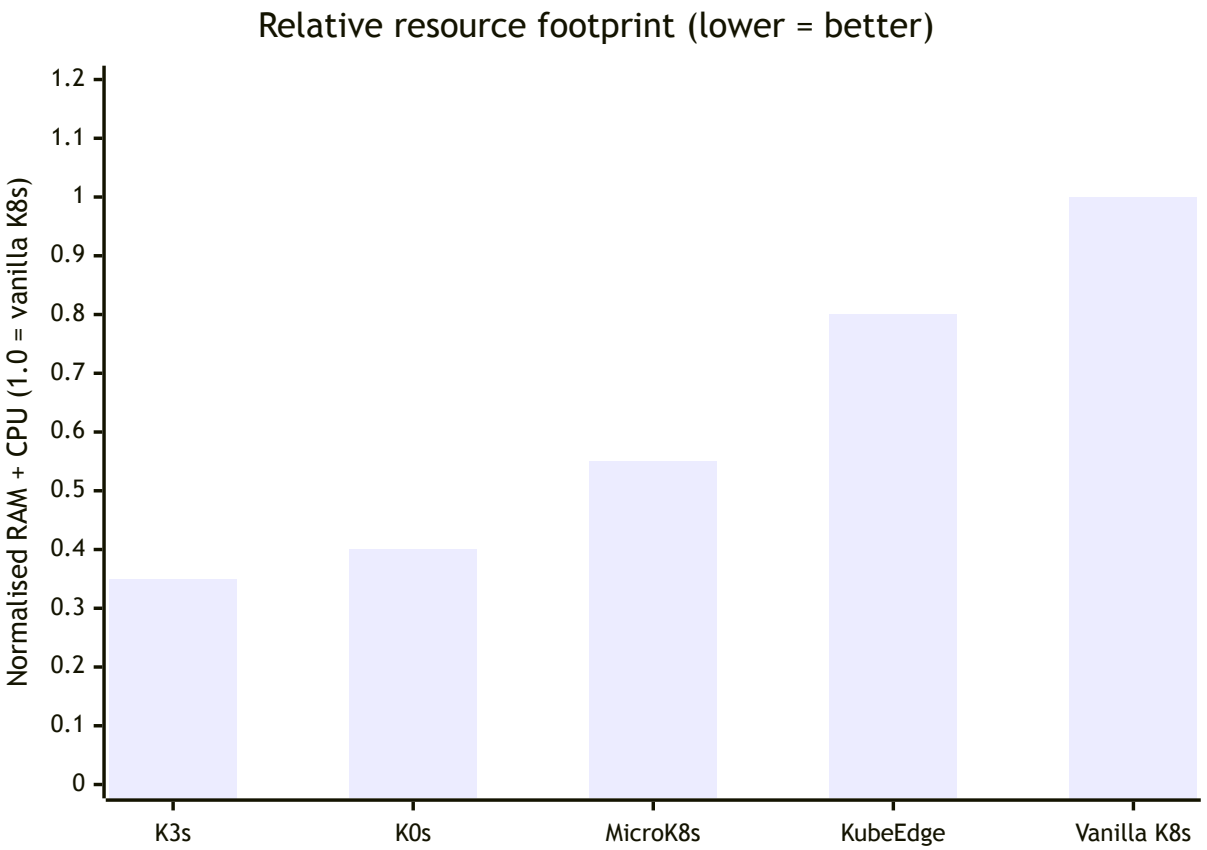

Figure 6. Approximate resource footprint of lightweight Kubernetes distributions relative to upstream Kubernetes, derived from the benchmarks in [13], [16].

4.2 Serverless at the Edge

Function-as-a-Service (FaaS) extends the edge toward event-driven, ephemeral execution. Cloudflare Workers use V8 isolates (~~5 MB / < 5 ms cold start); Fastly Compute uses WebAssembly on Wasmtime (~~35 µs); AWS Lambda@Edge layers regional edge functions onto CloudFront [17]. Open-source stacks such as OpenFaaS, Knative, and Kubeless allow self-hosted FaaS on edge Kubernetes clusters. The principal operational concern remains **cold-start latency**, surveyed comprehensively by Golec *et al.* in 2024 with a taxonomy of mitigation techniques (pool-warming, snapshot resume, pre-initialised runtimes) [17].

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5. Edge AI and Federated Learning

5.1 Edge Inference and Hardware Acceleration

Zhou *et al.*'s canonical survey divides the field into "AI *for* the edge" — using AI to optimise orchestration — and "AI *on* the edge" — running inference on edge hardware [19]. Murshed *et al.* provide an implementation-focused companion covering pruning, quantisation, distillation, and framework support (TF Lite, PyTorch Mobile, ONNX Runtime) on NVIDIA Jetson, Google Coral, and Intel Movidius [20]. A mature edge platform typically integrates a heterogeneous silicon mix — CPU for orchestration, GPU or NPU for vision inference, FPGA for deterministic real-time processing — choosing per workload rather than per rack.

5.2 Federated Learning

Federated learning (FL), introduced by McMahan *et al.* in 2017 as the FedAvg algorithm, trains global models without ever centralising raw data: nodes train locally and share only gradient updates with an aggregator [21]. Kairouz *et al.*'s 2021 agenda paper, with 58 co-authors, remains the canonical research reference [22]. Lim *et al.* bridged FL specifically to mobile edge networks [23], and Bonawitz *et al.* proved a practical secure-aggregation primitive enabling the aggregator to sum gradients without seeing individual contributions.

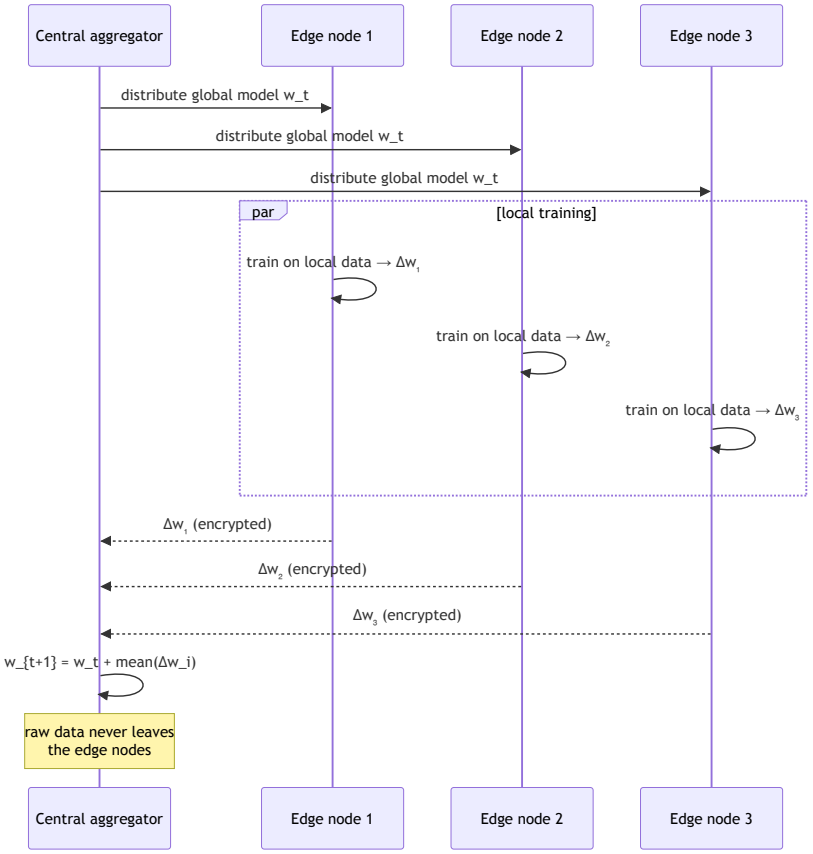

Figure 7. One round of federated averaging (FedAvg) [21].

5.3 Hierarchical FL and 6G

With the advent of 6G targeting massive IoT, FL is becoming hierarchical: edge devices form clusters, aggregate locally at a base station or fog node, and only then contribute to a global update [18]. This multi-tier design reduces communication overhead and addresses extreme device heterogeneity, allowing even ultra-low-power sensors to contribute meaningfully to a global model. Letaief *et al.*'s 2022 vision paper remains the canonical reference for the 6G + edge AI convergence [18].

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6. Security and Sovereignty

6.1 Zero-Trust Edge Architectures

Edge deployments dramatically expand the attack surface: nodes are often physically accessible and housed in uncontrolled environments. The **zero-trust (ZT) model** — "never trust, always verify" — is the foundational response [24], [25]. ZT mandates continuous per-session verification using contextual signals (identity, device posture, location), microsegmentation to contain lateral movement, and distributed policy enforcement points that can isolate a compromised node in sub-second timescales without waiting for a central SOC [24]. Sedjelmaci & Ansari (2023) propose an AI-augmented ZT framework tailored to 6G edge, using deep-learning-based behavioural analysis to detect anomalies at line rate [26].

6.2 Confidential Computing and Trusted Execution

Feng *et al.* (2024) comprehensively survey confidential computing primitives applicable to the edge: Intel SGX/TDX, AMD SEV, ARM TrustZone/CCA, and RISC-V Sanctum/Keystone [27]. A trusted execution environment (TEE) provides hardware-enforced confidentiality and integrity for code and data, enabling remote attestation so that a cloud or operator can verify an edge node's software state before provisioning secrets or workloads [27].

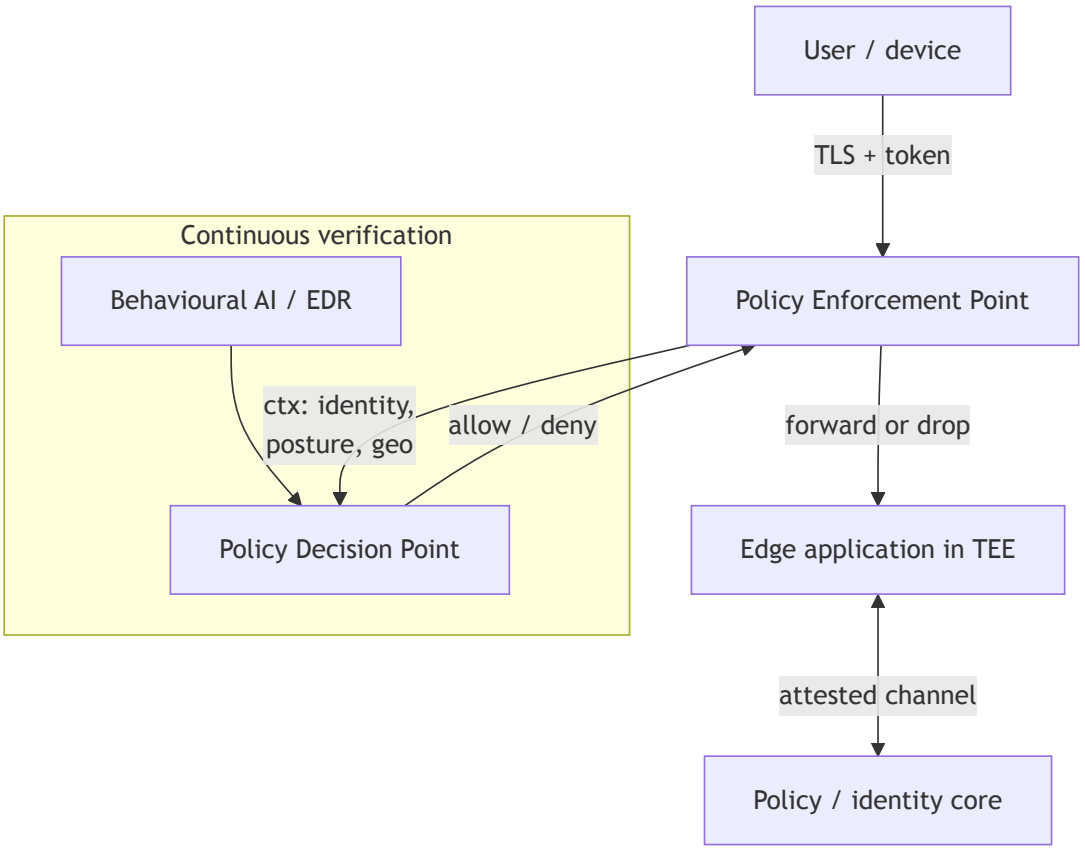

Figure 8. Zero-trust control loop at the edge with TEE-protected application [24], [26], [27].

6.3 Data Sovereignty

Residency regulations such as the EU GDPR and sector-specific rules (HIPAA, DORA) increasingly require that sensitive data remain within specified jurisdictions. Edge architectures enable the *partitioned multicloud*, *tiered hybrid*, and *air-gapped* sovereign patterns in which crown-jewel data is processed on local or on-premises nodes, while elastic front-ends or aggregate analytics run in public cloud. Hyperscalers have responded with dedicated sovereign

offerings (Microsoft Cloud for Sovereignty, AWS European Sovereign Cloud, Google Distributed Cloud air-gapped mode).

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7. Representative Use Cases

Edge computing's justification ultimately rests on specific workloads whose latency, bandwidth, privacy, or survivability requirements the public cloud cannot meet.

7.1 Autonomous Driving and Cooperative Perception

Liu *et al.*'s 2019 survey, with hundreds of citations, remains the standard reference for vehicular-edge stacks [28]. A connected vehicle offloads latency-critical perception to a roadside or base-station MEC host; Ma *et al.*'s 2020 5G-V2X field test demonstrated a full MEC-assisted prototype with real-time AI optimisation. V2X cooperative perception further enables vehicles to share their local detections through the MEC, expanding each vehicle's perceptual horizon beyond line of sight.

7.2 Industrial IoT and Industry 4.0

Sisinni *et al.* (2018) remains the foundational IIoT reference [29]. Industrial edge deployments combine Time-Sensitive Networking (TSN, IEEE 802.1AS/Qbv/CB/Qcc) for deterministic Ethernet transport, OPC UA for semantic interoperability, and fog-style hierarchical compute for control-loop latency guarantees. Modern factory floors integrate AI-based predictive maintenance and defect detection at the edge, uploading only flagged events to the cloud.

7.3 Smart Cities and Public Safety

Khan *et al.*'s 2020 comprehensive survey is the umbrella reference [30]. Real deployments include Liverpool NSW's Jetson TX2 + YOLOv3 video-analytics pipeline for traffic monitoring, various smart-grid edge intelligence schemes, and edge-augmented public-safety camera networks.

7.4 Immersive Media (AR / VR / Metaverse)

Mangiante *et al.* (2017) is the seminal edge-VR reference, addressing 360° video delivery in mobile networks [31]. Liu *et al.* (2019) demonstrated real-time edge-assisted object detection for mobile AR, and the emerging "edge-enabled metaverse" work formalises the MEC / metaverse convergence.

7.5 Healthcare

Edge architectures in healthcare must balance real-time bedside analytics with strict privacy regulation. Recent work combines fog-computing architectures with IoT-integrated deep learning (e.g., the FETCH framework) for patient monitoring, and federated learning has become a standard proposal for multi-institutional medical-image models that cannot share raw images due to privacy law.

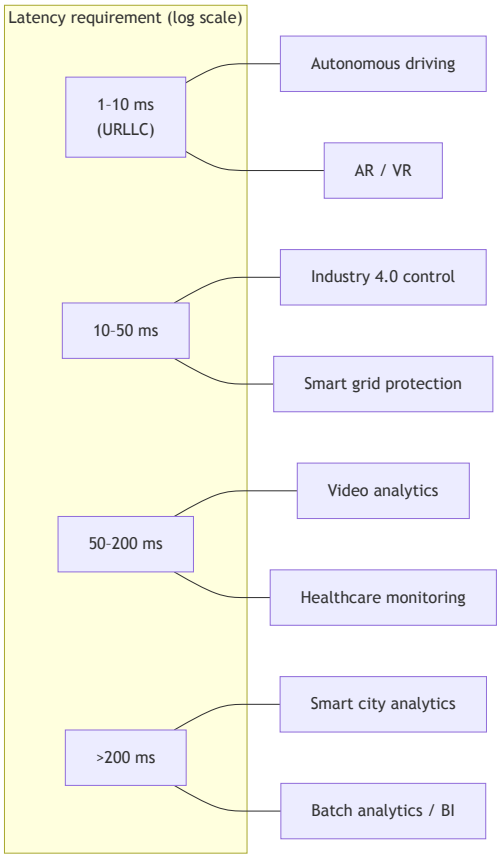

Figure 9. Latency profile of representative edge workloads.

8. Discussion and Future Outlook

Three observations summarise the 2022–2025 research frontier.

First, paradigm convergence. Cloudlets, fog, and MEC are no longer competing definitions. Satyanarayanan [2] and Ren *et al.* [10] treat them as deployment flavours of one continuum, and every major 2022–2025 survey — Kong 2023, Cruz 2022 [33] — builds on this premise.

Second, operational scale has replaced definitional debate. The dominant research themes are now Kubernetes-based orchestration for the edge [13], [15], serverless cold-start mitigation [17], 6G-native edge AI [18], and zero-trust / confidential computing [26], [27]. The field's open question is no longer "what is the edge?" but "how do we operate 10 000 edge sites reliably and securely?"

Third, implementation maturity now diverges by application domain. Autonomous vehicles and immersive media map cleanly onto the standardised MEC paradigm with ETSI/3GPP-compliant prototypes [28], [31], [32], while industrial IoT remains firmly anchored in the fog tradition with TSN and OPC UA [29]. Healthcare and smart cities draw from both [30].

The strongest framing for a modern seminar paper is therefore that **edge computing has entered a "cloud-native edge" era**, defined by portable Kubernetes workloads, ETSI-standardised APIs, pervasive AI, and zero-trust-by-design — a thesis supported by every major source surveyed here.

9. Conclusion

Edge computing is not a replacement for the cloud but a structural complement that addresses the physical and economic limits of hyperscale centralisation. The three classical paradigms — cloudlets, fog, and MEC — have converged into a unified end–edge–cloud continuum. Modern deployments are cloud-native, built on lightweight Kubernetes distributions and increasingly augmented by serverless runtimes. Edge AI and federated learning are moving intelligence *and* model training to the periphery, while zero-trust and confidential-computing primitives respond to the expanded attack surface. From autonomous driving to smart manufacturing to healthcare, the pattern is consistent: workloads whose latency, bandwidth, or privacy constraints the cloud cannot satisfy migrate to the edge, and the remaining workloads stay where elasticity is cheapest. Successful architectures therefore do not choose between cloud and edge — they place each workload on the right tier of the continuum.

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